

Saif Buheis

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EDUCATION

University of Toronto

Bachelor of Science in Computer Science, Geospatial Data Science, and Mathematics

Sep. 2022 – June 2026

Mississauga, ON

EXPERIENCE

ML Research Scientist

University of Toronto Mississauga

Jan. 2026 – Present

Mississauga, ON

- **8+ first-author manuscripts** (in preparation) under **Dr. Jenny Cui**, led by a **physics-informed ST-GNN** with compartmental mechanics that cut MAE **5.1%/8.8% at 2/4 weeks**, with **zero-shot** transfer to held-out states
- Built **soft-voting ensembles** (XGBoost, Random Forest, LightGBM) on **3.4M-row imbalanced** geospatial datasets over an **H3 hex grid**, using case-control sampling, probability calibration, and threshold optimization
- Designed a **deployment-grade validation suite** (OOD/temporal CV, **distribution-shift detection**, bootstrap CIs, **SHAP** rank-stability, **negative-control outcomes**) whose leakage guards caught **silent corruption in 36% of rows** and flipped a conclusion
- Contributed to a **diffusion-policy pipeline** at the **Medical Computer Vision and Robotics Lab**, deploying the policy on a **da Vinci surgical robot** for visuomotor manipulation

PROJECTS

Cadillac F1 Dashboard | XGBoost, Flask, React, PostgreSQL, Selenium | [Video Demo](#)

Feb. 2026 – Feb. 2026

- **Placed top 10%** at the Global Launch Challenge case competition as a team of 4 CS students competing against finance and business teams, presenting a full-stack operational dashboard to address Cadillac F1's 2026 launch risks
- Achieved **19x ROI** on predictive maintenance by building a Gradient Boosting ML model (**$R^2 = 0.91$**) that forecasts the Remaining Useful Life of 8 F1 car components, **enabling £2.85M in savings** per critical intervention
- Built a real-time brand sentiment monitoring system by integrating scraped X and Reddit data into themed dashboard panels, giving the communications team visibility into public narrative across platforms

Resume Roaster | Next.js 15, Python, AWS Bedrock, Lambda, Docker, RAG | [Video Demo](#)

Nov. 2025 – Jan. 2026

- Launched a RAG application processing **1000+ resumes** with a 100% success rate via [amicooked.ca](#).
- Reduced latency by **35%** to bypass AWS timeout limits via **Claude 3.5 Sonnet** and **RAG** parameter tuning.
- Architected a serverless backend using **FAISS** and **Lambda**, cutting costs to **\$0.011 per roast**.
- Deployed a containerized **Python** pipeline using **Docker** to automate document chunking and embedding.

ML-Powered Bill Categorizer | Python, FastAPI, Angular, PostgreSQL, Scikit-learn

Aug. 2025 – Dec. 2025

- Hit **96% accuracy** on **5,000+ simulated transactions** by training **SGDClassifier** on 1,000+ labeled records.
- Designed **PostgreSQL** schema for incremental retraining based on real-time user feedback loops.
- Developed **ETL pipelines** via **NumPy/Pandas** for vectorized preprocessing and low-latency inference.

Roof Damage Detection | PyTorch, OpenCV, GDAL, ArcGIS Pro, Python

Jul. 2025 – Aug. 2025

- Engineered a **U-Net segmentation model** in **PyTorch** to quantify roof damage from drone imagery.
- Boosted **IoU** by **12%** via an **ETL pipeline** with **OpenCV/GDAL** to neutralize shadows and image artifacts.
- Improved detection of rare damage patterns by applying **data augmentation** and a **weighted loss function**.
- Developed a **sliding-window tiling strategy** to process large GeoTIFFs and calculate damage area in m^2 .

LEADERSHIP

Shift Supervisor

Skyzone

Mar. 2024 – Present

Mississauga, ON

- Awarded **Sales Member of the Year** for both **2024 and 2025**, recognizing consistent top-tier performance.
- Generated **\$11,115 in monthly revenue** by achieving the highest volume of \$195 annual pass sales.

TECHNICAL SKILLS

Languages: Python, Java, SQL (Postgres), TypeScript/JavaScript, R

ML & Data: PyTorch, Scikit-learn, XGBoost, LightGBM, SHAP, NumPy, Pandas, OpenCV, LangGraph

Geospatial & Tools: GeoPandas, OSMnx, H3, GDAL, ArcGIS Pro, Docker, AWS (Lambda, Bedrock), Git